National University of Computer & Emerging Sciences, Karachi
Fall-2017 CS-Department
Lab Final Exam

Course Code: EL-213	**Course Name:** Computer Organization & Assembly Lang Lab
Instructor Name(s): Syed Zain ul Hassan, Sehrish Saeed, Sumaiyah Zahid	
Student Roll No:	**Section No: G**
Date: 5-12-2017	**Time: 1:30 - 3:30**

Instructions:
Attempt all tasks
Return the question paper after exam

Max Marks: 20 points

Task1: (5 marks)

You are given a sequence of 22 uninitialized bytes starting from variable *myVar*. Now consider the following table:

Byte#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Op	4	5	7	9	8	3	5	6	8	0	9	7	4	2	4	6	7	8	9	7	5	4

Write a program which first fills the 22 bytes with any random number between 0 to 15 and then lets user select a byte. Then rotate the selected byte according to corresponding ***op*** given in the table for that byte. In addition to that, each even byte will rotate left and odd byte will rotate right.

Task 2: (5 marks)

Write a program which counts the frequency of each character with other and display it in well-defined format.

String: “Coal is love.”

Task 3: (5 marks)

Write a program for an ATM machine which contains procedures named as *CheckPassword, Deposit, Withdrawal and BalanceInquiry.* The ATM will authenticate any user if the previous set password got matched otherwise the card will be blocked. There must be nested function calling between *CheckPassword* and the rest one. Then from within *CheckPassword*, call other functions if and if password is correct in 3 trials.

In every function call other than *CheckPassword* you have to display the address of the runtime stack of its calling procedure (*CheckPassword*).

Task 4: (5 marks)

Write a program which contains a procedure to read characters from a keyboard until the last four characters read make up the word D O R Y. Display the number of characters that were read. (Read 25 characters at max). Then encrypt the total input string to next to next character.

Sample Output: (Assuming character sequence A F G D O R Y were entered)

Characters Read: 7